Universidad de La Laguna

Continuous Integration (CI), Vite, Github Actions and Github pages.

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Link to code examples in these slides

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What is Continuous integration?

Let’s see an example without continuous integration

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Let's see an example without continuous integration

Developer A

Developer B

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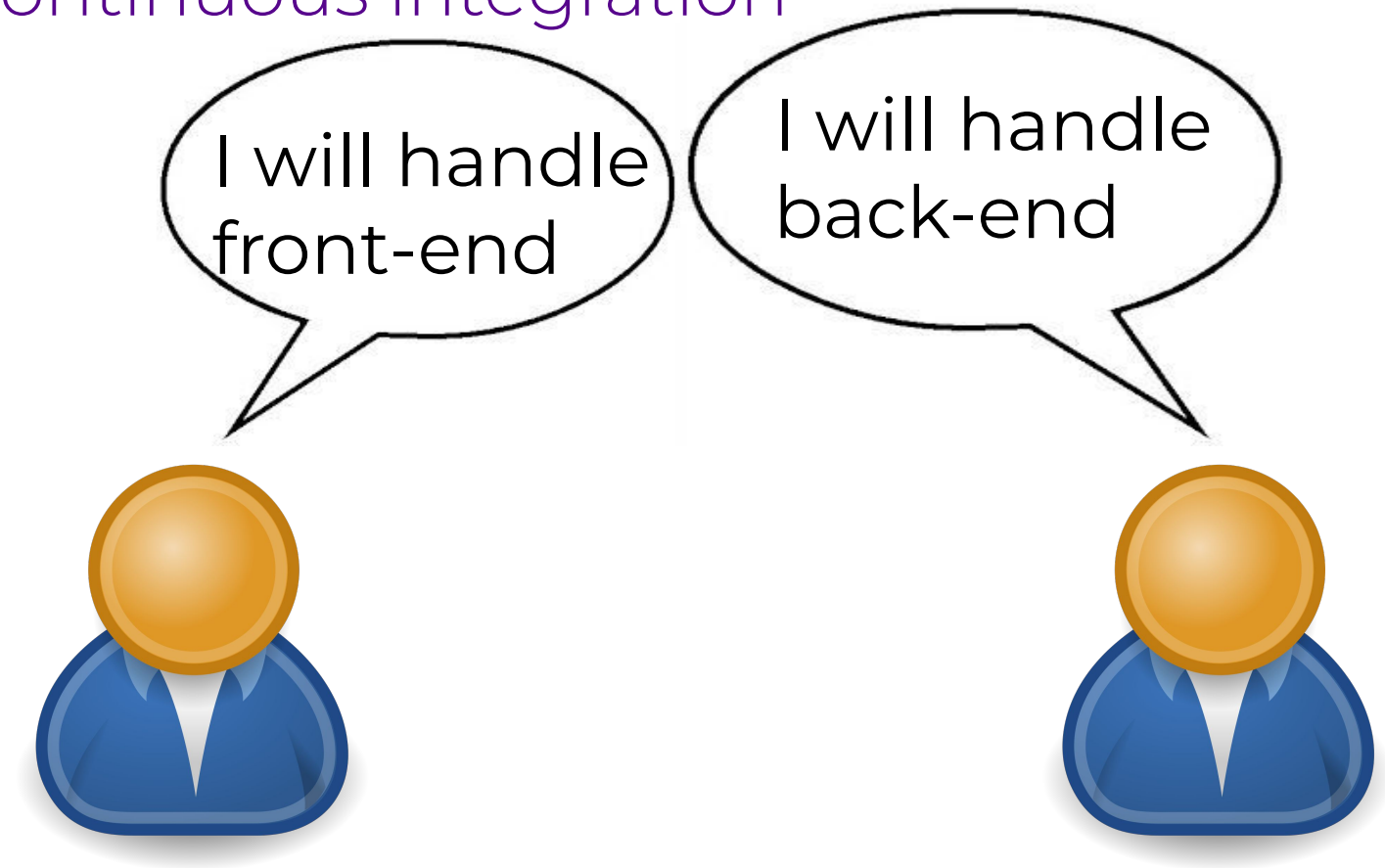
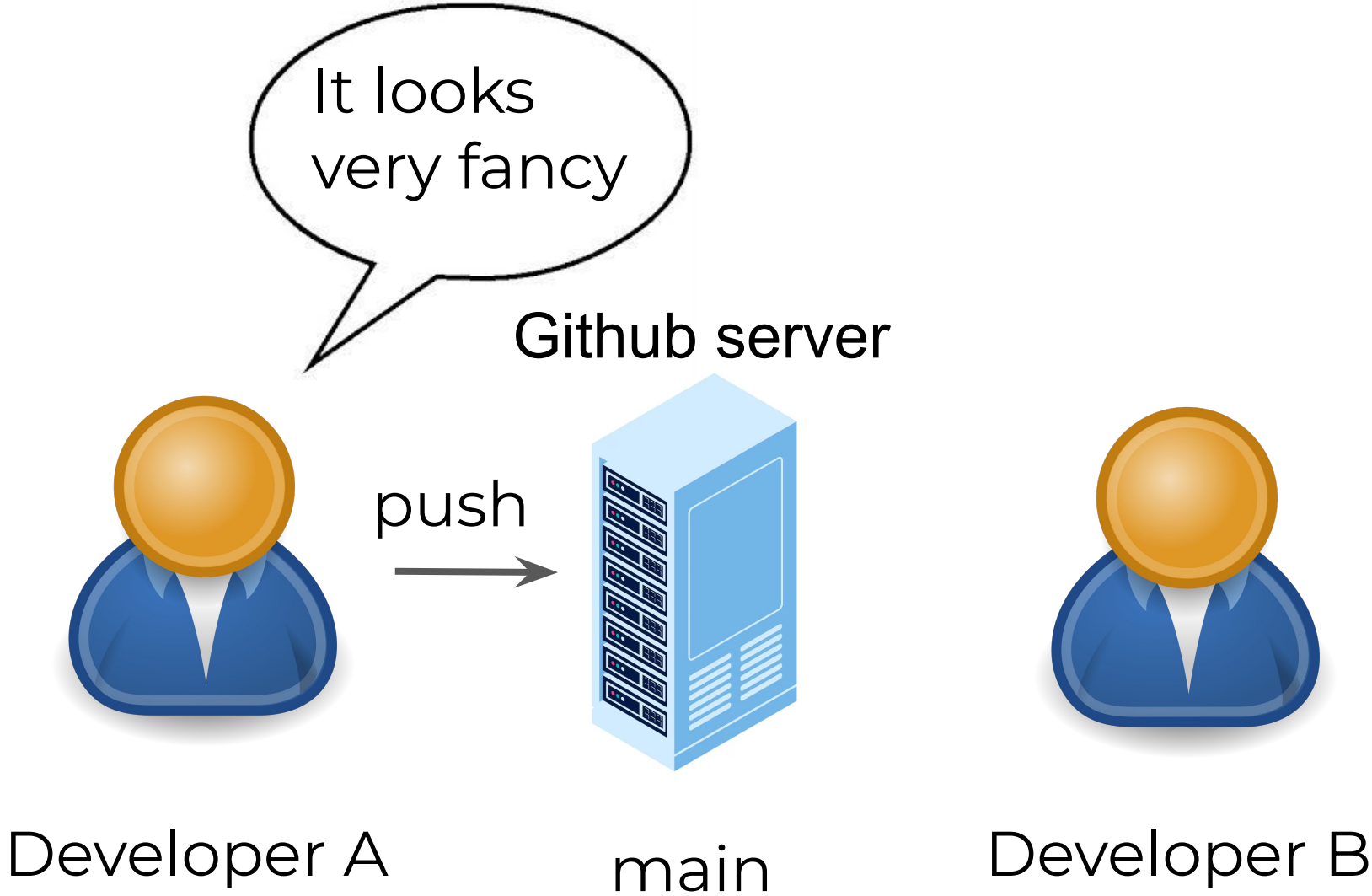
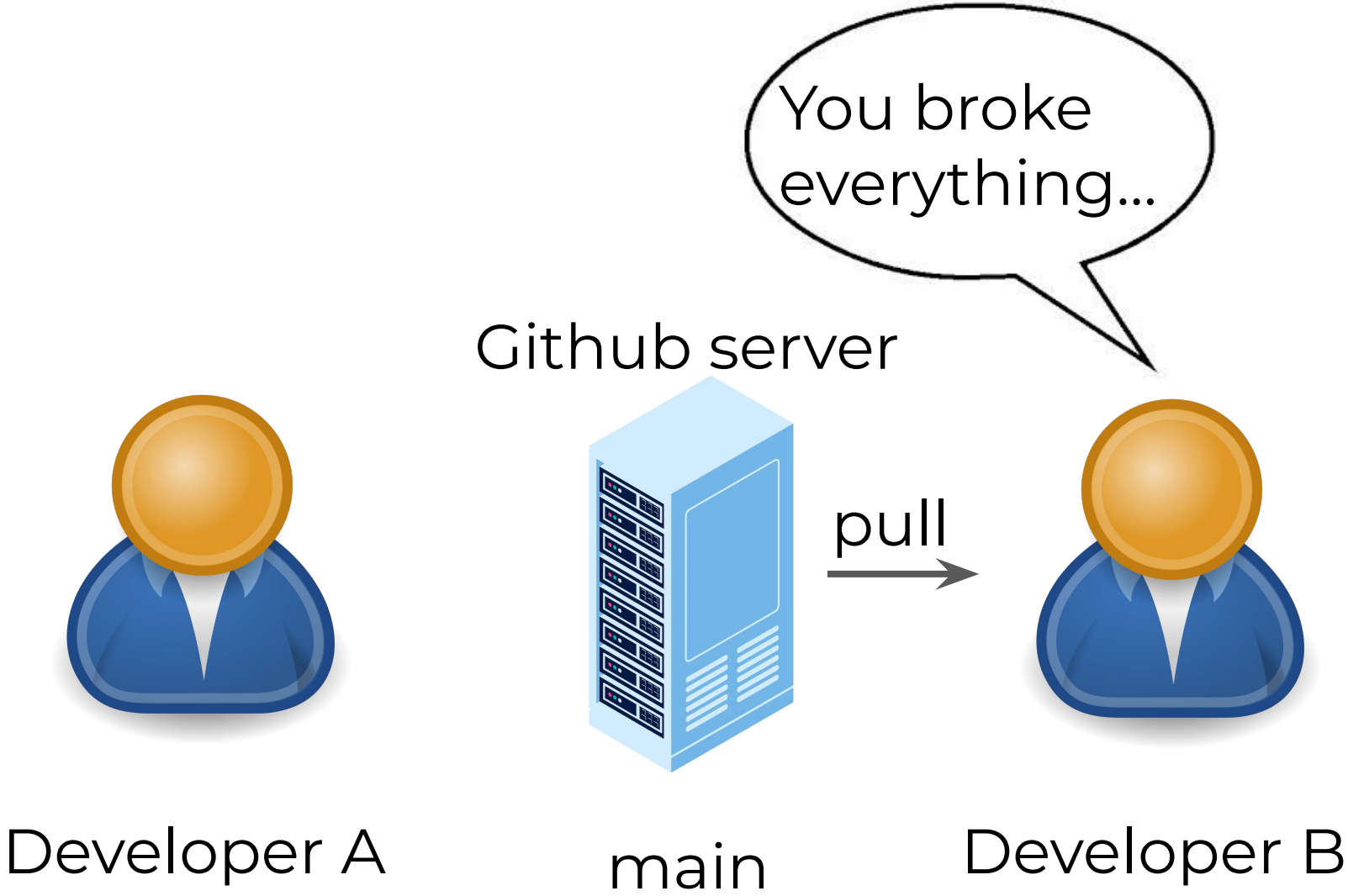
Let's see an example without continuous integration




It looks very fancy
Github server
push
Developer A
main
Developer B


You broke everything...
Github server
pull
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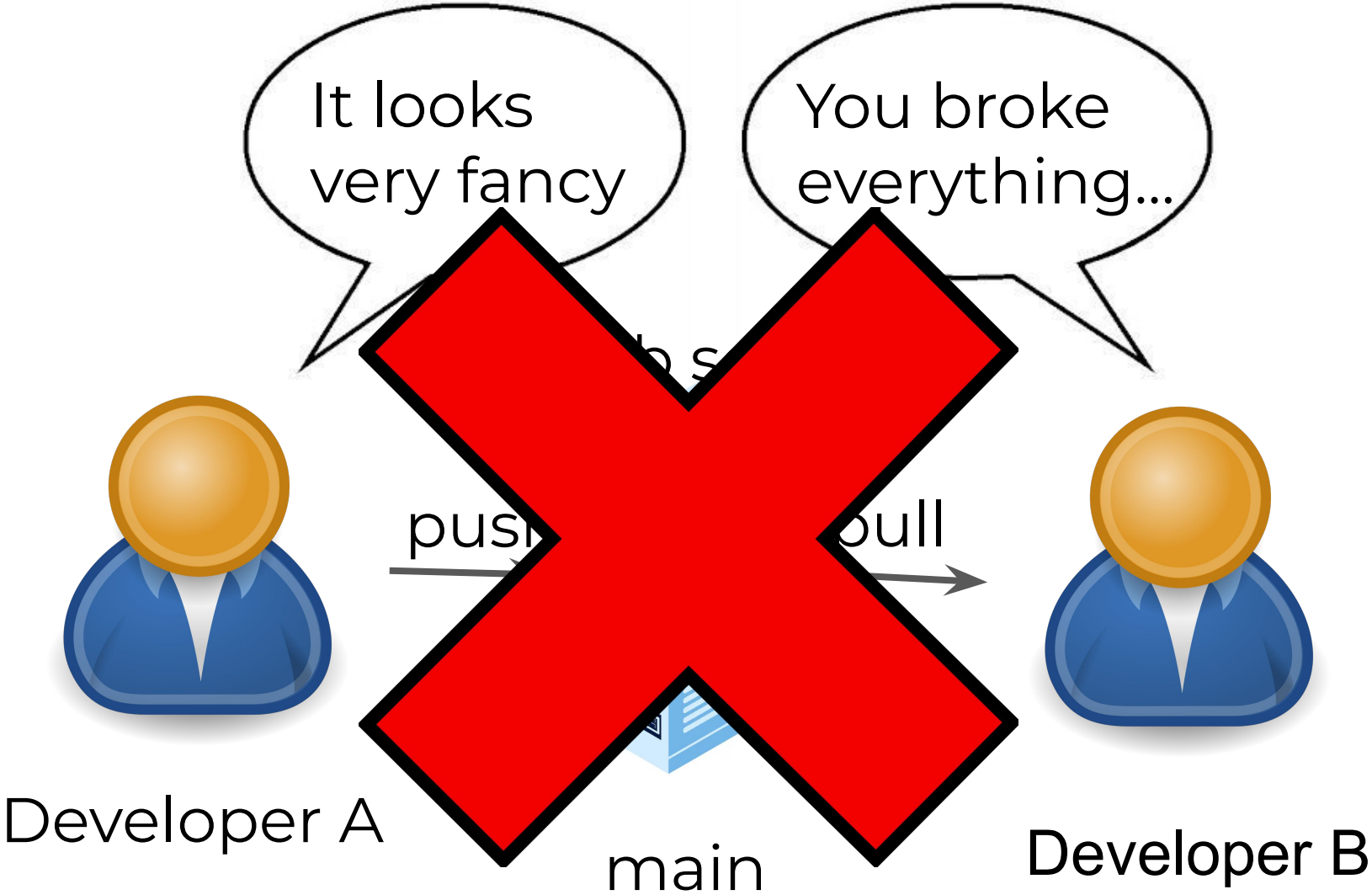
main
Developer B


It looks very fancy
You broke everything...
Developer A
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How do we avoid that?

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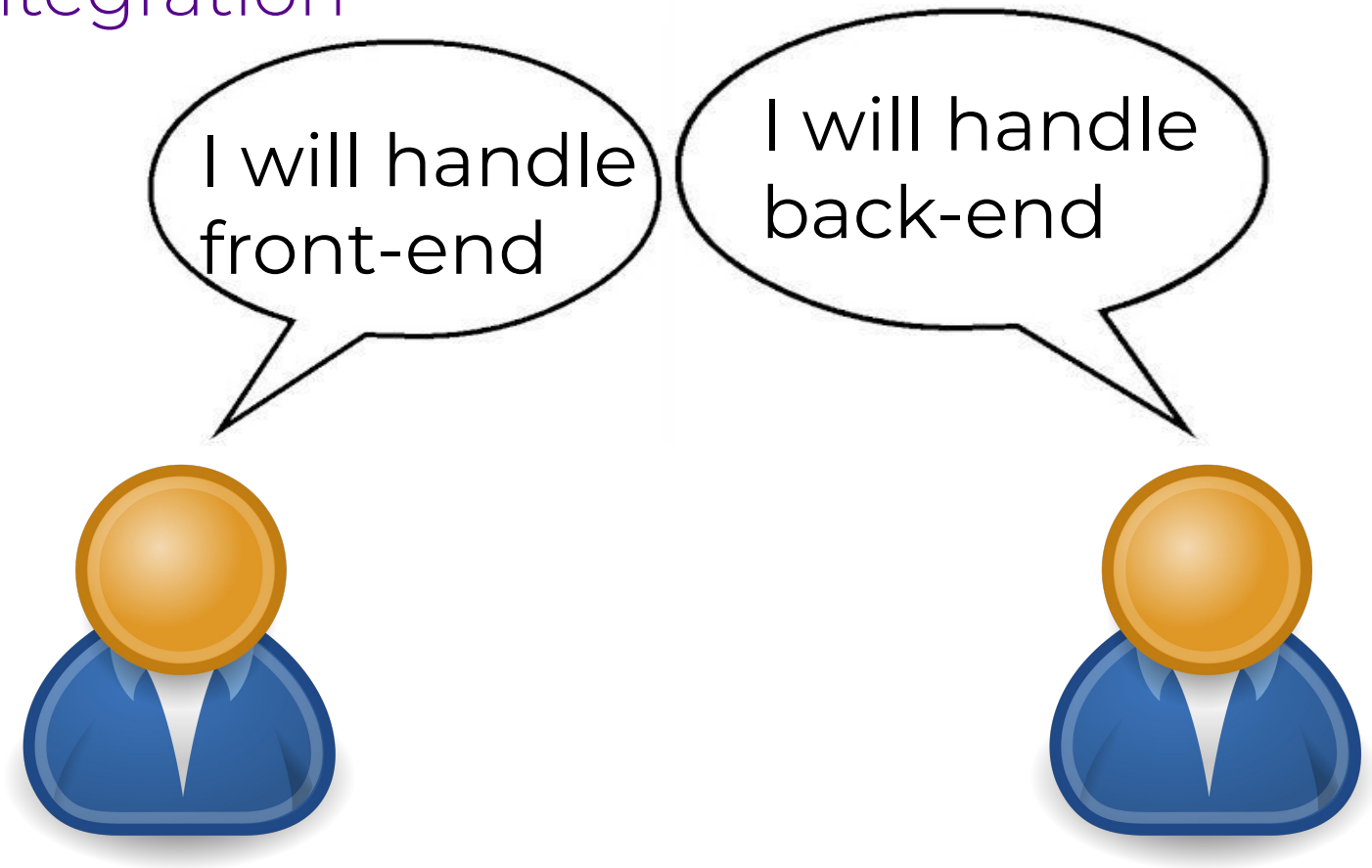
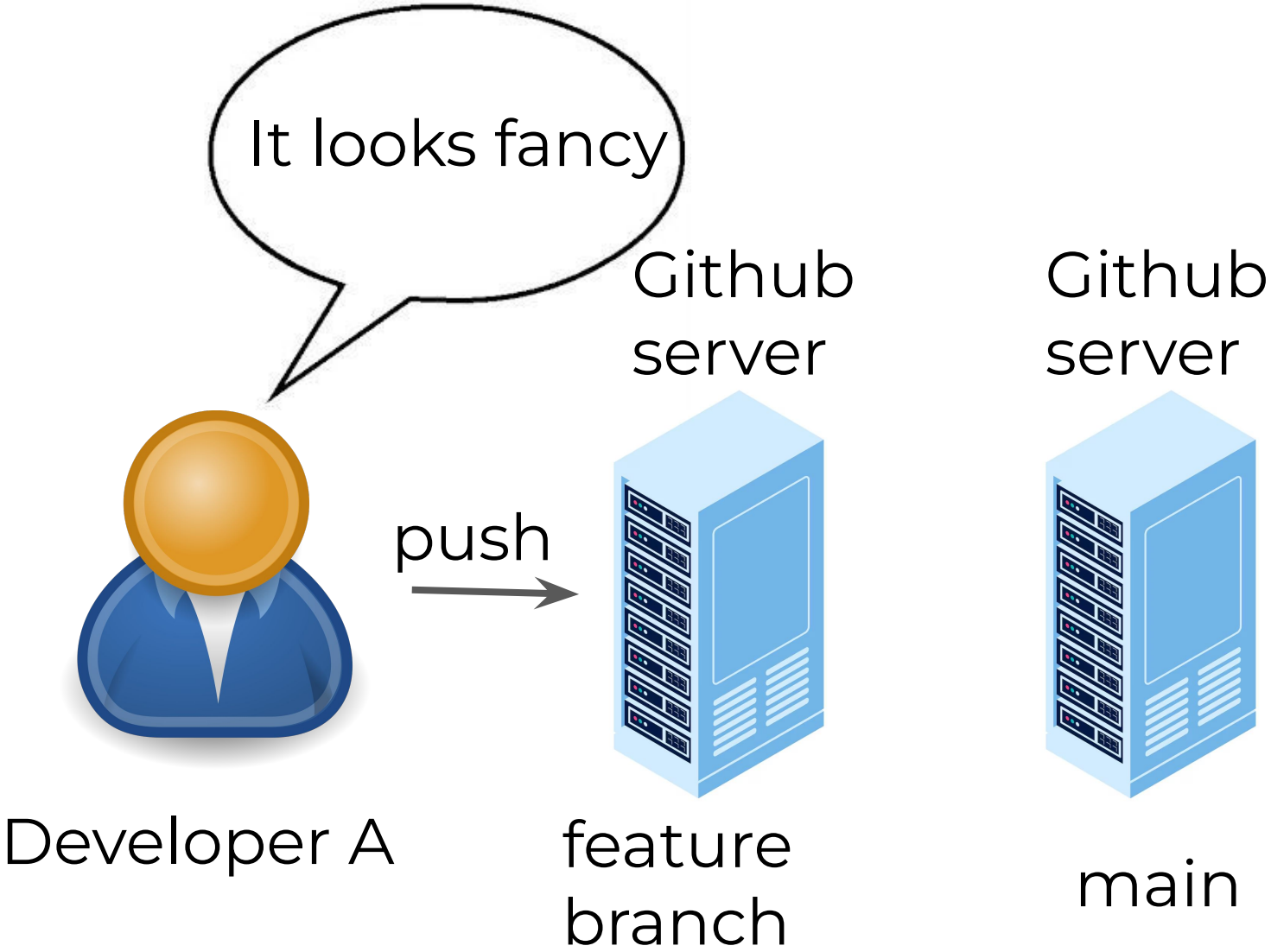
Continuous integration

What is Continuous integration?

Let's see an example with continuous integration




It looks fancy
Github server
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feature branch
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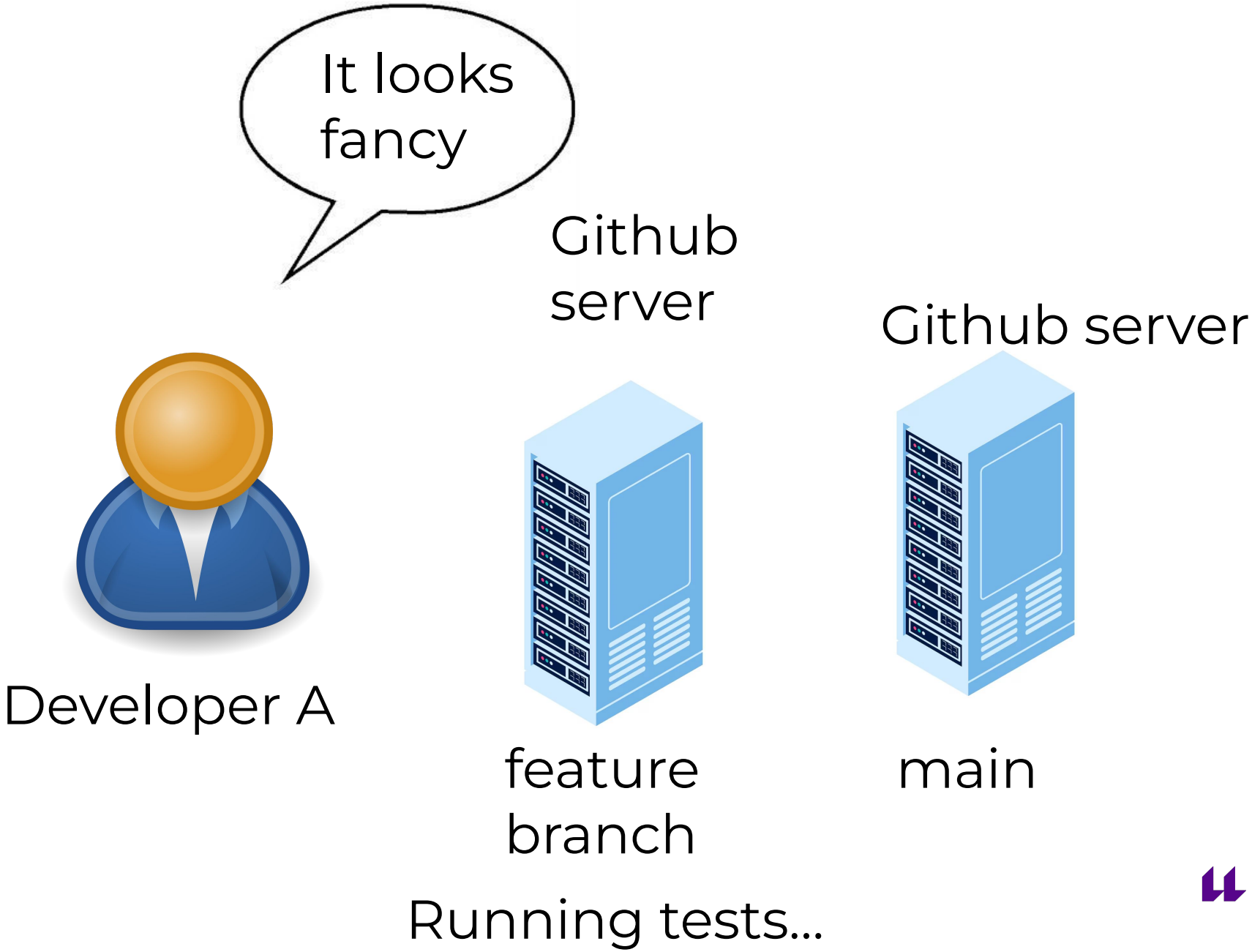
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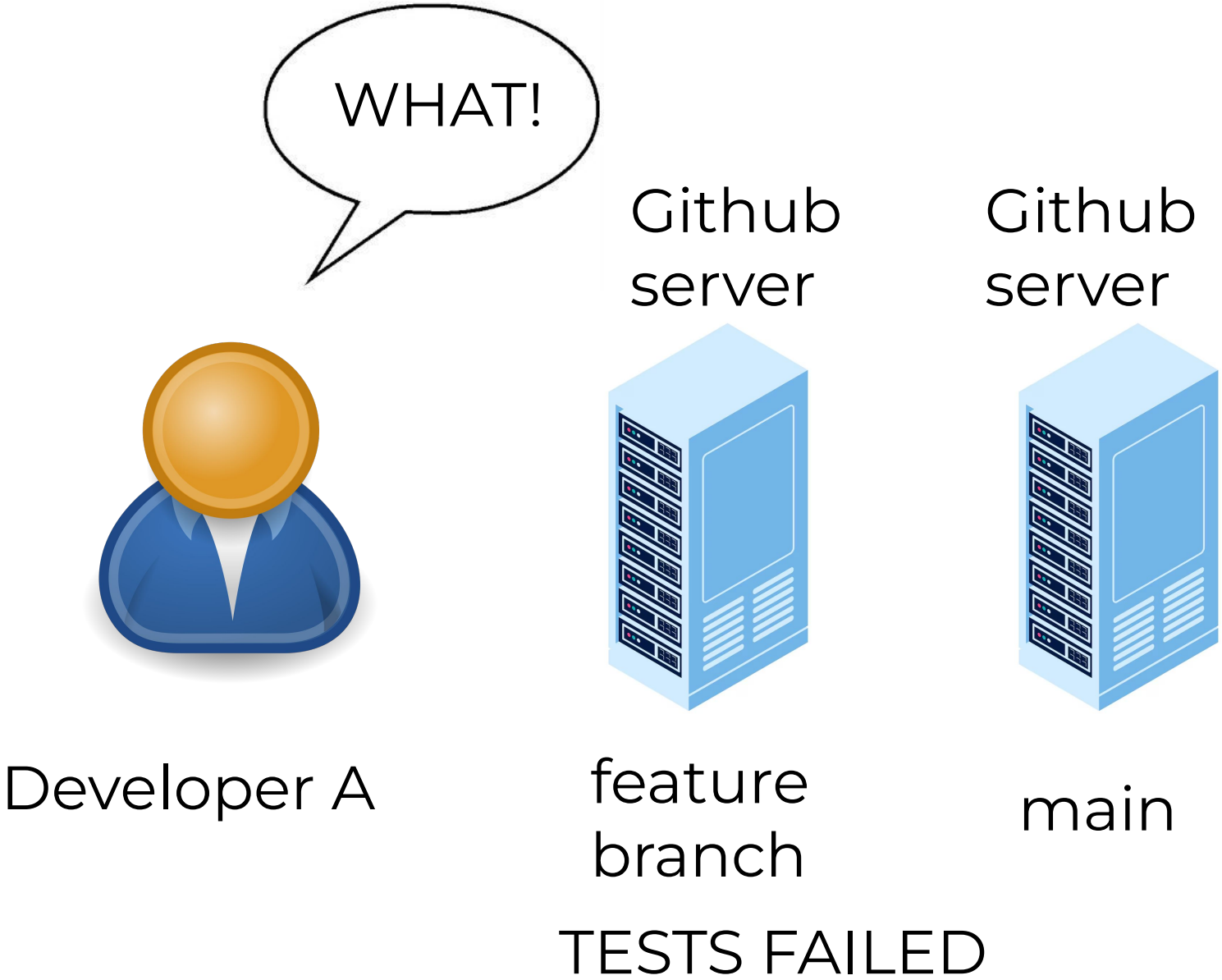
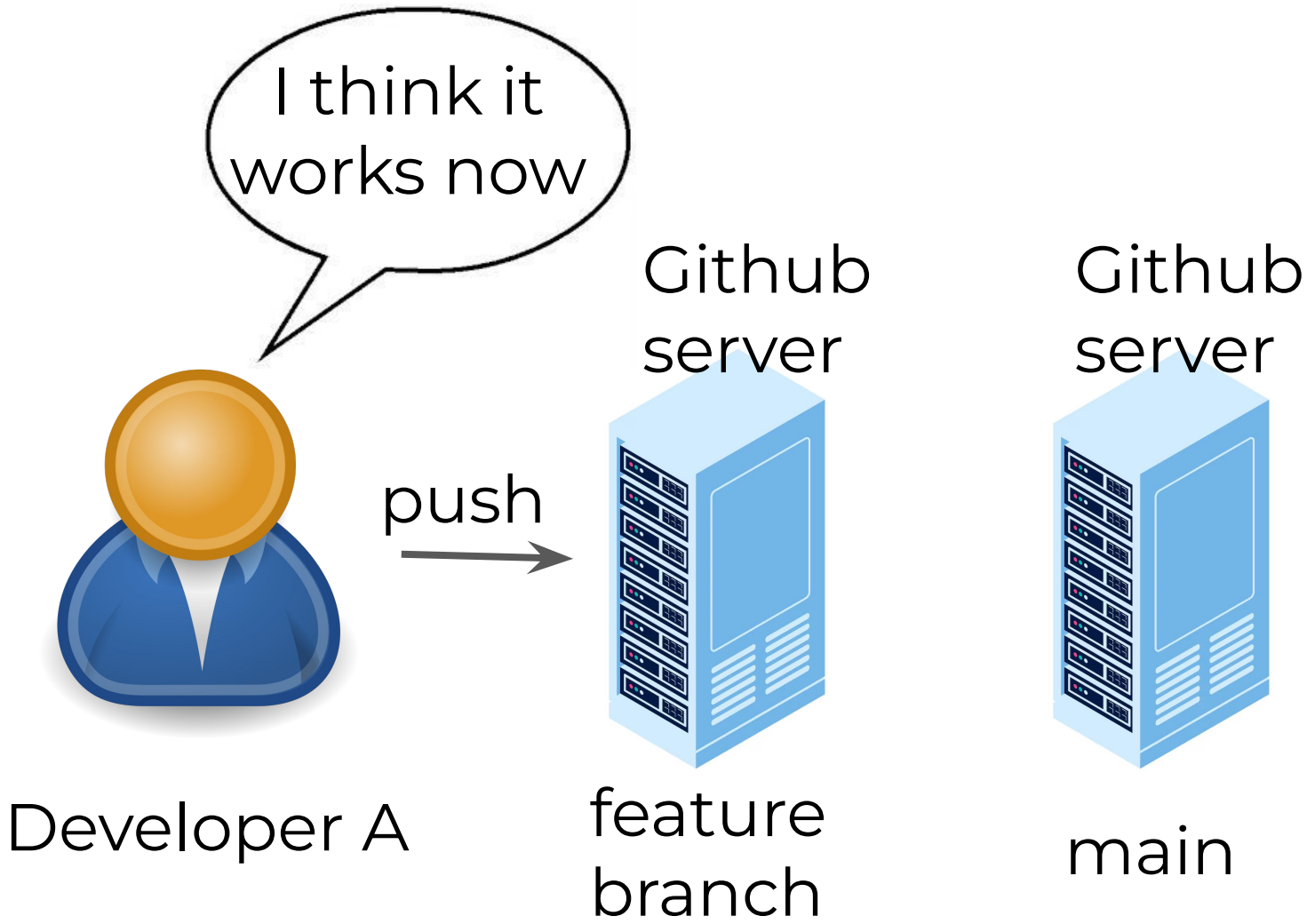
Running tests...

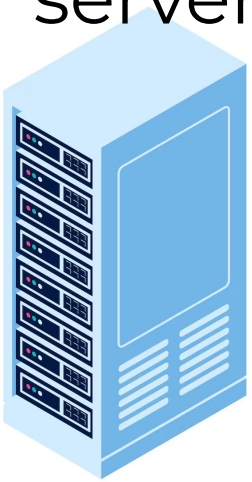

WHAT!
Github server
Github server
Developer A
feature branch
main
TESTS FAILED



Running tests...

Developer A

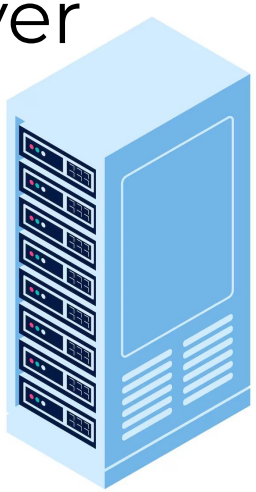
Github server



feature branch

merge

Github server



main

Tests succesful

Developer A

Developer B

Vite

- Modern frontend build tool
- Focused on **speed and simplicity**
- Great fit for TypeScript projects

Why use Vite?

Problems with Traditional Tools:

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- Slow startup time

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Problems with Traditional Tools:

- Slow startup time
- Full rebundling on every change
- Poor scalability in large projects

Key Features of Vite

- Instant server start

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- Fast Hot Module Replacement (HMR)

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- Instant server start
- Fast Hot Module Replacement (HMR)
- Native ES Modules support
- Optimized production build
- Simple and minimal configuration

How to install

```
$ npm create vite@latest
$ npm install
$ npm run dev
```

Optional configuration file: `vite.config.ts`

The `index.html` file must be in the root of the project.

Github Pages

- Some key characteristics:
  - Static website hosting service

Github Pages

- Some key characteristics:
  - Static website hosting service
  - Directly from a GitHub repository

Github Pages

- Some key characteristics:
  - Static website hosting service
  - Directly from a GitHub repository
  - Free and easy to use

Github Pages

- How does it work?
  - Push code to repository
  - GitHub selects a branch
  - Static files are served as a website

Github Pages

- How does it work?
  - Push code to repository
  - GitHub selects a branch
  - Static files are served as a website

- Deployment process with Vite
  - Build project locally (`npm run build`)
  - Generate static files (dist/)
  - Upload files to repository branch

Let's see an example: Repository

What do we have to change?

- Change `index.html` as follows:
  - Before:

```html
<!-- Debe haber un index.html en la raíz del proyecto -->
<link rel="stylesheet" href="./styles.css">
```

```html
<!-- Enlaces relativos a dist/ -->
<a href="../src/exercises/home-work/projectile.html" class="btn">Ver simulación</a>
```

  - After:

```html
<!-- Debe haber un index.html en la raíz del proyecto -->
<link rel="stylesheet" href="./public/styles.css">
```

```html
<!-- Enlaces relativos a dist/ -->
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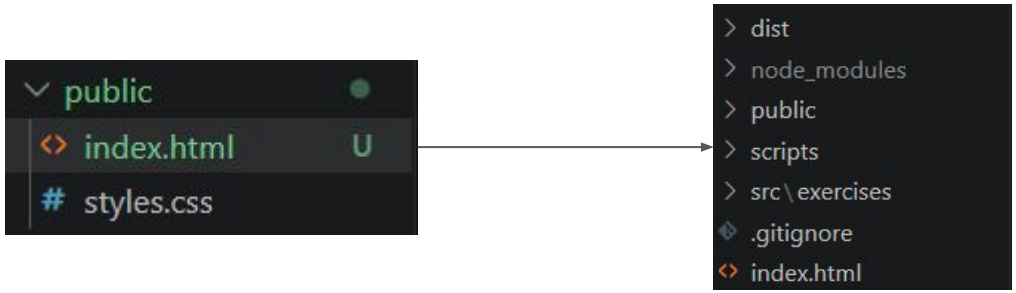
```

  - **Remember**: `index.html` in root directory



What do we have to change?

- Configure `vite.config.ts`:
  - Base: <RepositoryName>/dist/

```
base: '/2025-2026-pai-integraci-n-continua-2025-2026-pai-integracion-continua/dist/',
```

  - Input: Integrate **ALL** HTML

```
input: {
  index: resolve(__dirname, './index.html'),
  p10Projectile: resolve(__dirname, './src/exercises/home-work/projectile.html'),
  p10UML: resolve(__dirname, './src/exercises/home-work/uml/projectile-uml.html'),
  p10GraphicObject: resolve(__dirname, './src/exercises/evaluation-exercise1-title/graphic-object.html'),
}
```

What do we have to change?

- Change `package.json` dependences:

```
"vite": "^8.0.8",
"@types/node": "^20.19.0",
```

- Change `typedoc` output directory to lay in dist/:

```
"out": "dist/src/exercises/home-work/doc",
```

Study this example for further understanding: [gh-pages-fundamentals]

Github Pages

- Github Pages are usually combined with Github Actions, allowing automatic build and/or deployment.

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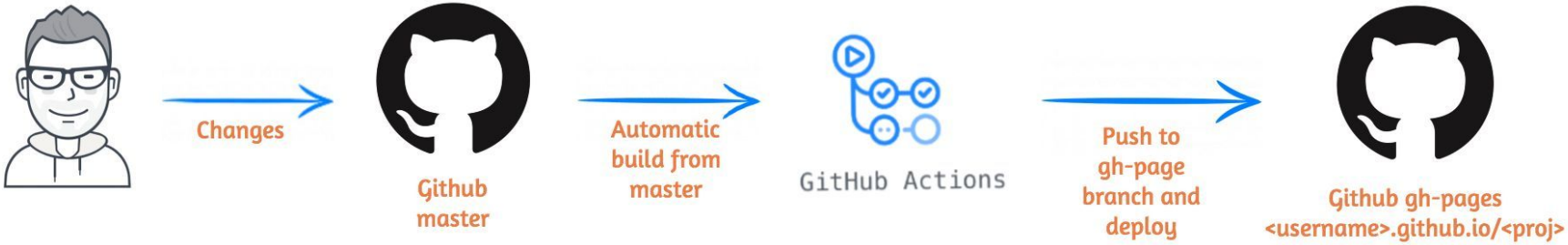


Github Actions

- Automation
- Integrated
- Event-driven

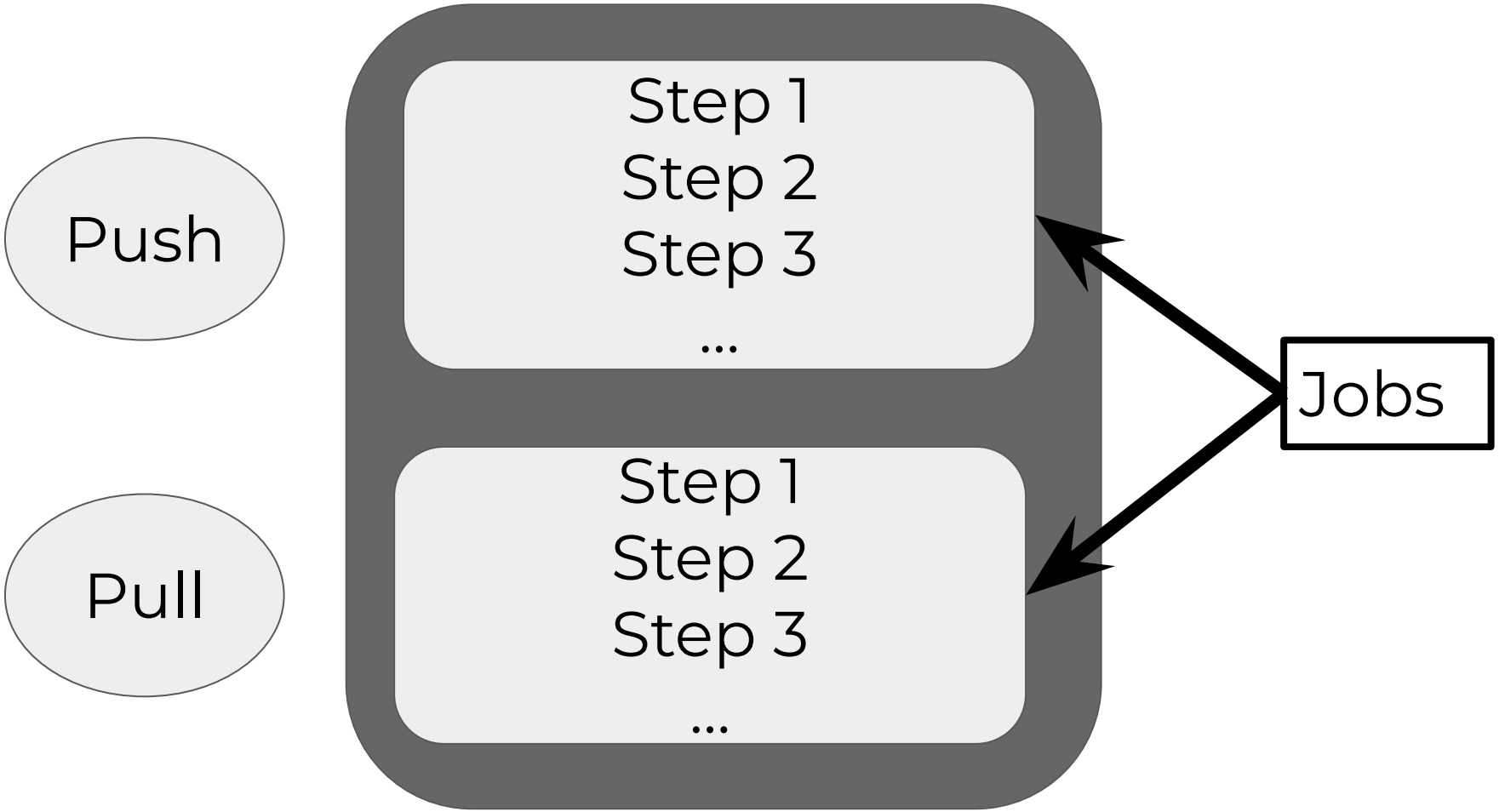
Github actions

- To use GitHub Actions, you need one or more YAML files.

EVERY `.yml` is evaluated automatically inside `.github/workflows`

- triggered by certain events(pull, push, etc...)

Github actions



Github actions

What do we want to add inside `.github/workflows`?

We are going to create a workflow that:

- Automatically runs our code
- Installs project dependencies
- Executes tests
- Detects errors before merging

General structure of a workflow

Main keys:

- name
- on
- jobs
- uses
- runs-on
- steps

```
name: CI Pipeline
on: [push, pull_request]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - run: npm install
       - run: npm test
```

“main” and “on” keys

`name`: Name of the workflow

`on`: When the workflow runs:

- `push`
- `pull_request`
- `workflow_dispatch`

“jobs” and “uses” keys

`jobs`: Defines the set of independent or interconnected tasks.

`uses`: Runs a pre-made action created by GitHub, the community, or your team.

“runs-on” and “steps” keys

`runs-on`: Where the job runs (Ubuntu, Windows, or macOS)

`steps`: Defines the sequence of individual commands or actions that execute in order within a job.

Github Actions Examples:
github-actions-fundamentals

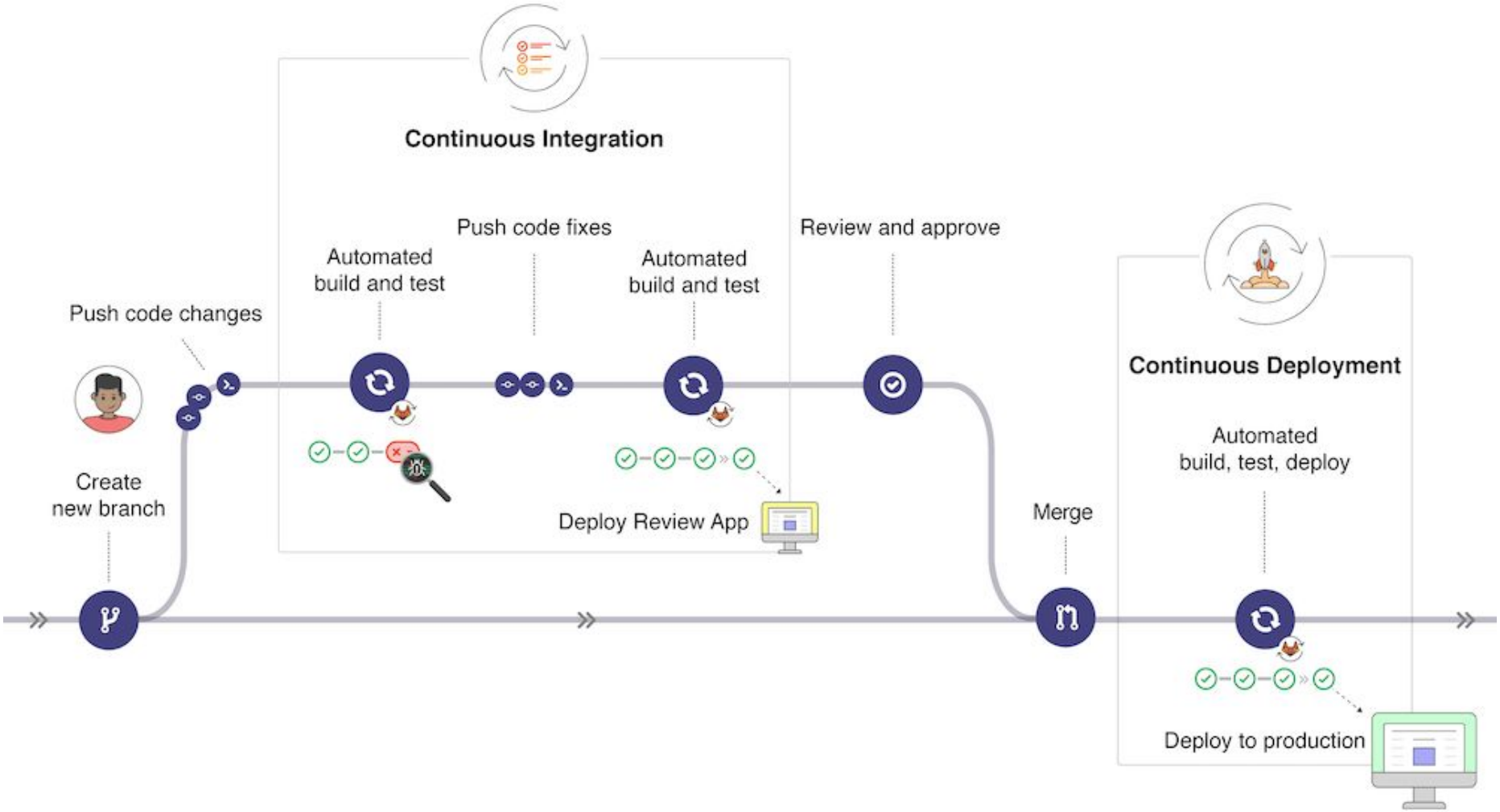
Continuous Delivery

- Code is always in a deployable state
- Build, test, deploy to staging (all automated)
- Production deployment requires a human click
- QA team validates before going live

Continuous Deployment

- Ship features to users as fast as possible
- 100% automated - no manual steps
- Production deployment happens automatically
- Users get updates immediately

Continuous Deployment



Monorepos

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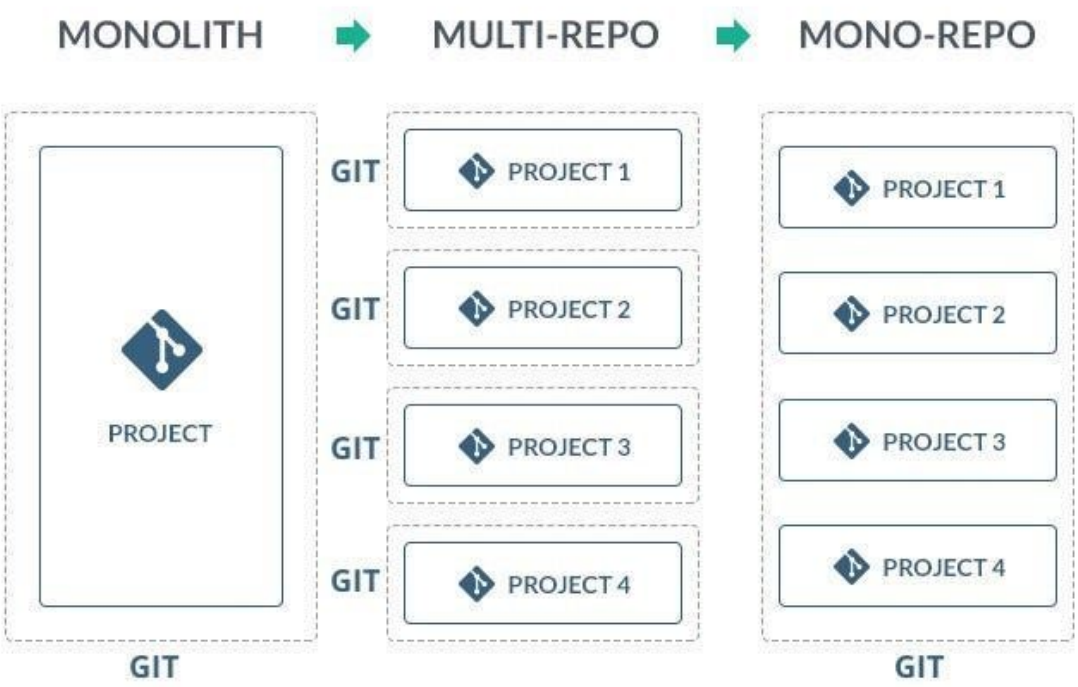

Monorepos

- Create a repository where the monorepo will be allocated in.

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$ git init
$ mkdir src
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```
$ git remote add <name> <HTTP link>
$ git fetch —-all
$ git subtree add --prefix=src/<directory-name> <name>
main —-squash
```

Monorepos

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$ mkdir src
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- Add the repositories with:

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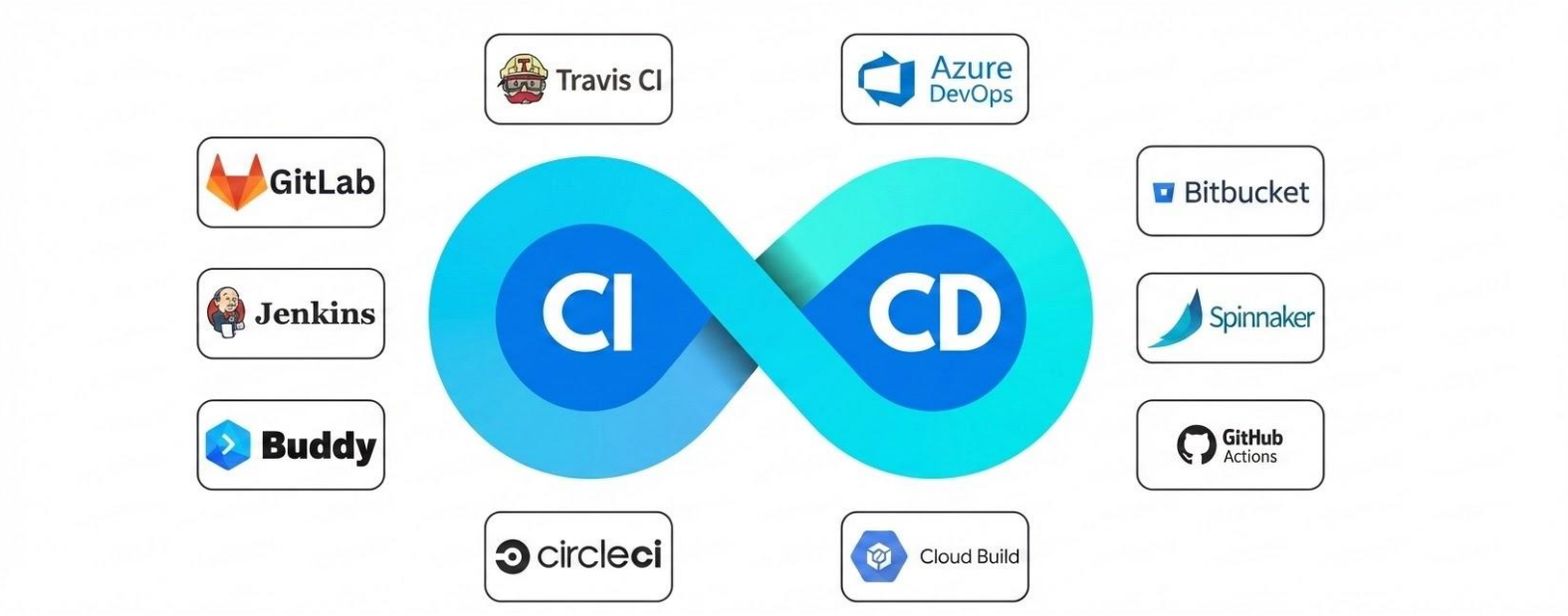
$ git subtree add --prefix=src/<directory-name> <name>
main —-squash
```

- Configure `package.json` and `vite.config.ts`

Let’s see an example: PAI-Monorepo

CI/CD Alternatives

- Some popular alternatives to GitHub Actions:
  - Jenkins
  - GitLab CI/CD
  - CircleCI
  - Google Cloud Run



CI/CD Alternatives - Jenkins

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Jenkins

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- Problems
  - Complex setup

Jenkins

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- Some key characteristics of Jenkins:
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  - Highly customizable (plugins)
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- Problems
  - Complex setup
  - Requires maintenance

Jenkins

CI/CD Alternatives - GitLab CI/CD

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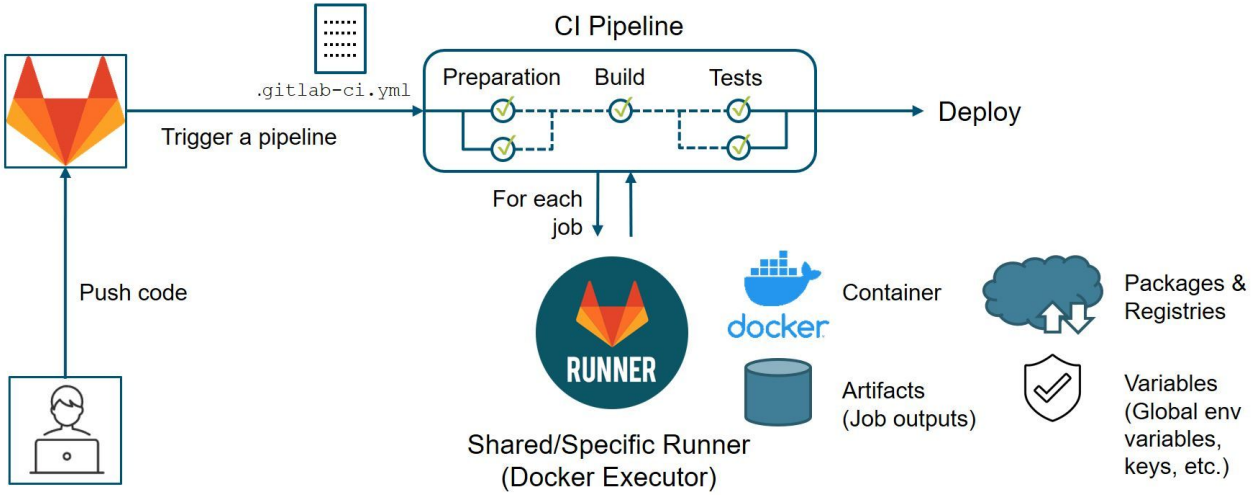
  - All-in-one platform
  - Easy integration



CI/CD Alternatives - CircleCI

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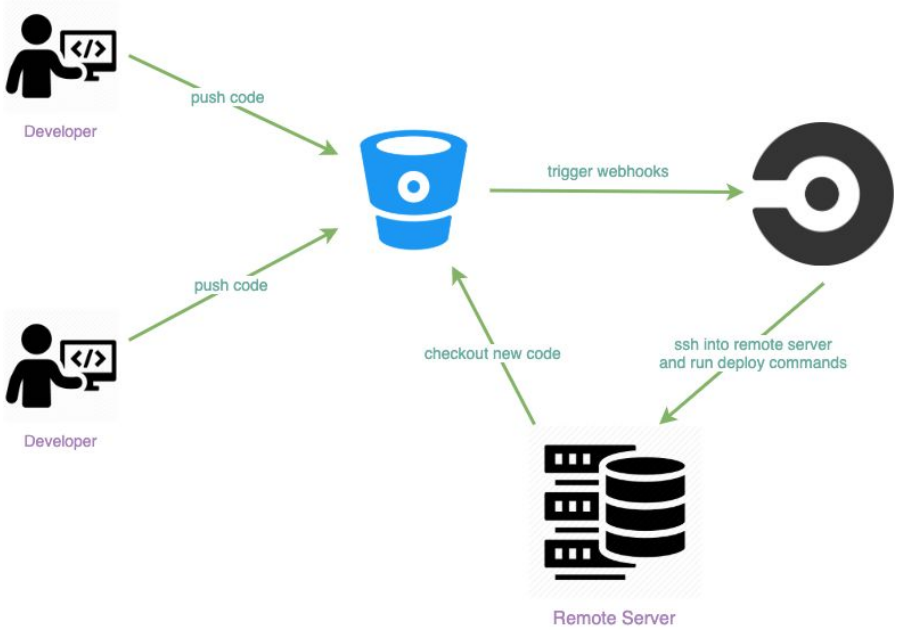
- Advantages
  - Easy to use
  - High performance



CI/CD Alternatives - Google Cloud

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CI/CD Alternatives - Google Cloud

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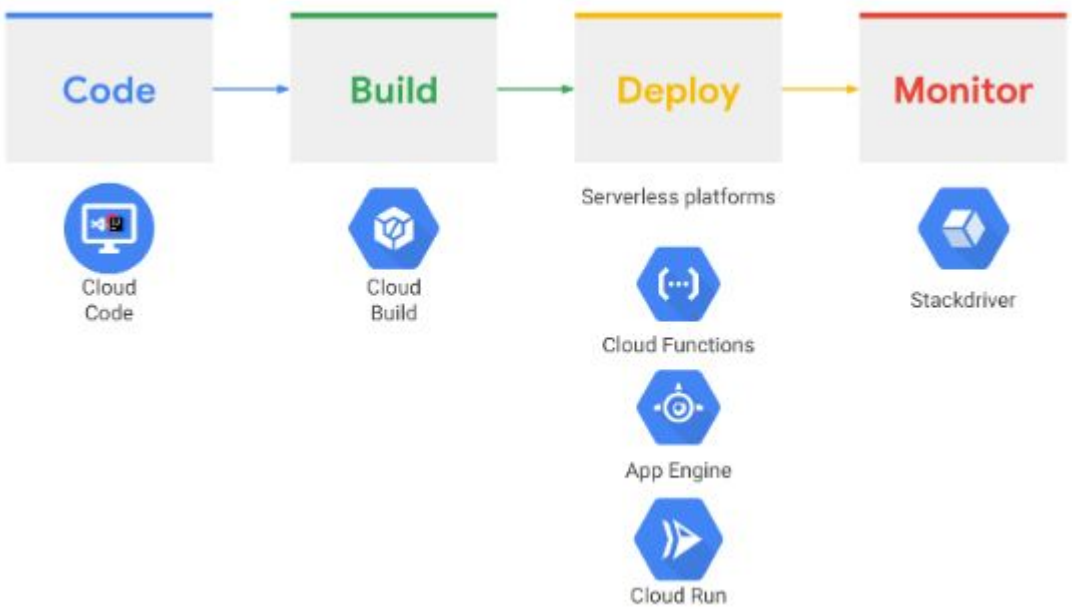
CI/CD Alternatives - Google Cloud

- Some key characteristics:
  - Paid cloud platform
  - Runs containerized applications
  - Used for deployment, not only CI

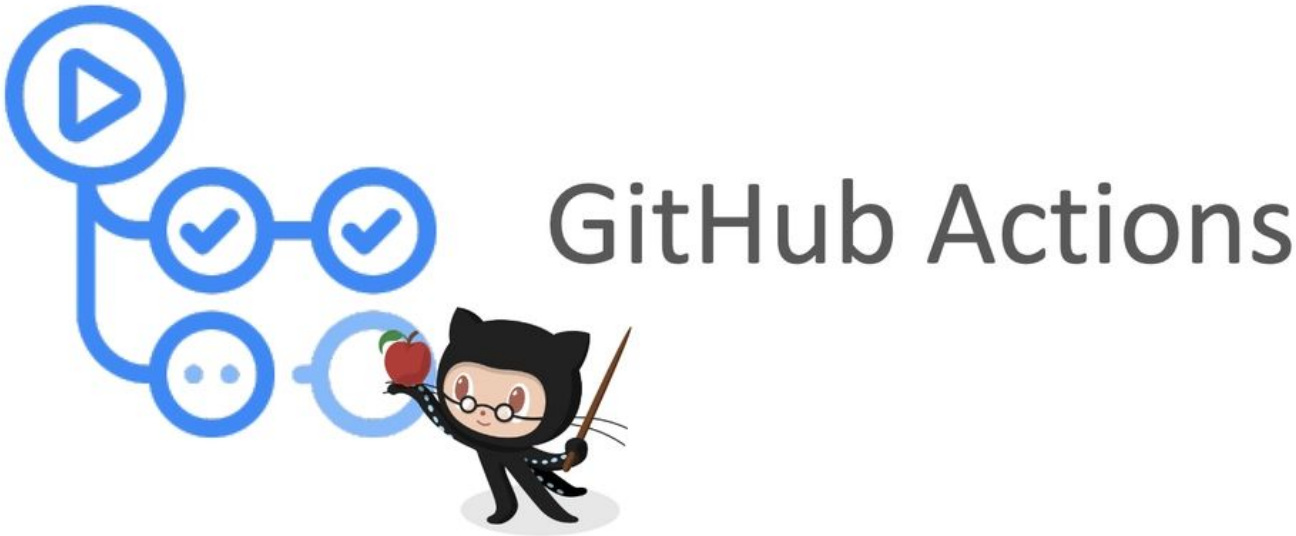


CI/CD Alternatives - Comparison

Tool	Type	Setup	Flexibility
Github Actions	Cloud	Easy	Medium
Jenkins	Self-hosted	Hard	High
GitLab CI/CD	Integrated	Medium	High
CircleCI	Cloud	Easy	Medium
Google Cloud	Cloud (Paid)	Medium	High

So, why GitHub Actions?

- Native GitHub integration
- No installation required
- Free tier available
- Simple workflows with YAML



References

- Learn Continuous Integration and Delivery with GitHub Actions, Docker, and Google Cloud Run(1 hour)
- GitHub Actions(10 minutes)
- GitHub Actions TUTORIAL (15 minutes)
- Setting up a TypeScript project with Vite(5 minutes)
- Step-by-Step Guide: Deploy Your Vite Project to GitHub Pages with gh-pages (5 minutes)
- Vite website(5 minutes)

Any question?

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